Beckn Protocol Extension Schemas

Domain extension specifications for the NFH-Kuza-Google rural service delivery pilot

DOMAIN CODE	PROTOCOL VERSION	EXTENSION SCHEMAS	SUB-SCHEMAS
`local-services`	Beckn v2.0.0	7 schemas	4 shared types

This document defines the domain extension schemas for a **local-services** domain within Beckn v2.0.0. The domain enables rural households to discover and transact with local service providers across any service category through a single window, using WhatsApp as the consumer-side BAP interface and a web app as the provider-side BPP interface.

> **Sub-categories are dynamic.** Within each category, sub-categories are managed as master data in the Kuza Platform and are not hardcoded in these schemas. The `subCategory` field uses a generic `{code, name}` structure to support any sub-categorization.

Category Codes

`Livelihood-Services`
LIVELIHOOD FLOW
Any income/livelihood-related service or input (veterinary, crop, trades/skills services)

`Household-Services`
HOUSEHOLD FLOW
Any household need (education, health, insurance, consumer goods)

`Market-Produce`
MARKET FLOW
Aggregation and sale of any goods/produce. Farmer is the seller, provider is the buyer/aggregator.

`Advisory-Content`
ADVISORY FLOW
Any informational/training content delivery. Free, discover-only — no order lifecycle.

CATEGORY CODE	FLOW	HAS ORDER LIFECYCLE	PAYMENT
`Livelihood-Services`	Livelihood	Yes	Paid
`Household-Services`	Household	Yes	Paid
`Market-Produce`	Market	Yes	Paid
`Advisory-Content`	Advisory	No — discover only	Free

Beckn v2 Core Entity Mapping

These extension schemas attach to Beckn v2 core entities through the designated extension containers:

EXTENSION CONTAINER	BECKN V2 CORE ENTITY	SCHEMAS	PURPOSE
`itemAttributes`	Resource / Item	LivelihoodItemAttributes, HouseholdItemAttributes, MarketProduceItemAttributes, AdvisoryContentItemAttributes	Describes what the service/item is and what the consumer needs
`offerAttributes`	Offer	LocalServiceOfferAttributes	Pricing, payment methods, availability, and capacity
`fulfillmentAttributes`	Performance / Fulfillment	LocalServiceFulfillmentAttributes	Location, scheduling, delivery mode, and service status tracking
`orderAttributes`	Contract / Order	LocalServiceOrderAttributes	Payment confirmation, service confirmation, and feedback from both parties

Schema selection is driven by category. A Beckn message for `Livelihood-Services` would carry `LivelihoodItemAttributes` in `itemAttributes`, `LocalServiceOfferAttributes` in `offerAttributes`, `LocalServiceFulfillmentAttributes` in `fulfillmentAttributes`, and `LocalServiceOrderAttributes` in `orderAttributes`. Advisory uses only itemAttributes and fulfillmentAttributes (for location context).

1. LivelihoodItemAttributes

lia: LivelihoodItemAttributes

Container: **itemAttributes** Applies to: **Livelihood-Services**

Describes a livelihood-related service or input that a farmer is seeking. Covers veterinary services, crop services, input supply, and trades/skills.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code, name}	REQUIRED	Domain grouping within Livelihood	`{"code": "livestock-services", "name": "Livestock Services"}`
`serviceName`	`string`	REQUIRED	Human-readable name of the service	`"Poultry Vaccination"`
`serviceDescription`	`string`	OPTIONAL	Detailed description of the service	`"Newcastle Disease vaccination for broiler chickens"`
`serviceMode`	`enum`	REQUIRED	How the service is delivered. Values: ON_SITE_VISIT PROVIDER_LOCATION INPUT_DELIVERY	`"ON_SITE_VISIT"`
`quantityRequired`	`QuantitySpec`	OPTIONAL	What the farmer needs serviced/supplied	`{"value": 200, "unit": "birds", "description": "Birds to vaccinate"}`
`estimatedDuration`	`string` ISO 8601	OPTIONAL	Estimated time to deliver the service	`"PT2H"`
`inputDetails`	`object`	OPTIONAL	For input supply: {inputName, unit, quantityPerUnit}	`{"inputName": "Broiler Feed", "unit": "Kg", "quantityPerUnit": 50}`
`sectorContext`	`object`	OPTIONAL	Domain-specific context: {livestockType, breed, cropType, tradeType}	`{"livestockType": "Poultry", "breed": "Broiler"}`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 language codes	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags for discoverability	`["vaccination", "poultry", "broiler"]`

OpenAPI 3.1 Definition

```yaml
# LivelihoodItemAttributes (lia:)
LivelihoodItemAttributes:
  type: object
  required:
    - subCategory
    - serviceName
    - serviceMode
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
        code:
          type: string
          description: Domain grouping code within Livelihood
```

```yaml
          examples: ["livestock-services", "crop-services", "trades-skills"]
        name:
          type: string
          description: Human-readable sub-category name
    serviceName:
      type: string
      description: Human-readable name of the service
    serviceDescription:
      type: string
      description: Detailed description of the service
    serviceMode:
      type: string
      enum: [ON_SITE_VISIT, PROVIDER_LOCATION, INPUT_DELIVERY]
      description: How the service is delivered
    quantityRequired:
      $ref: "#/components/schemas/QuantitySpec"
    estimatedDuration:
      type: string
      format: duration
      description: ISO 8601 duration
    inputDetails:
      type: object
      properties:
        inputName:
          type: string
        unit:
          type: string
        quantityPerUnit:
          type: number
    sectorContext:
      type: object
      properties:
        livestockType:
          type: string
        breed:
          type: string
        cropType:
          type: string
        tradeType:
          type: string
    availableLanguages:
      type: array
      items:
        type: string
        pattern: "^[a-z]{2}$"
    tags:
      type: array
      items:
        type: string
```

JSON Example — Beckn Message Fragment

```json
{
  "context": {
    "domain": "local-services",
    "action": "on_discover",
    "version": "2.0.0"
  },
  "message": {
    "catalog": {
      "providers": [{
        "id": "provider-uuid-001",
        "descriptor": { "name": "Dr. Wanjiku Veterinary Services" },
        "items": [{
          "id": "item-uuid-001",
          "descriptor": { "name": "Poultry Vaccination" },
          "category_ids": ["Livelihood-Services"],
          "itemAttributes": {
            "subCategory": { "code": "livestock-services", "name": "Livestock Services" },
            "serviceName": "Poultry Vaccination",
            "serviceDescription": "Newcastle Disease vaccination for broiler chickens",
            "serviceMode": "ON_SITE_VISIT",
            "quantityRequired": { "value": 200, "unit": "birds", "description": "Birds to vaccinate" },
```

```
            "estimatedDuration": "PT2H",
            "sectorContext": { "livestockType": "Poultry", "breed": "Broiler" },
            "availableLanguages": ["en", "sw"],
            "tags": ["vaccination", "poultry", "newcastle"]
          }
        }]
      }]
    }
  }
}
```

2. HouseholdItemAttributes

hia: **HouseholdItemAttributes**

Container: **itemAttributes** Applies to: **Household-Services**

Describes a household-level need: education, health, insurance, consumer goods, or any non-livelihood service.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code, name}	REQUIRED	Domain grouping within Household	`{"code": "education", "name": "Education"}`
`serviceName`	`string`	REQUIRED	Human-readable name	`"School Enrollment Assistance"`
`serviceDescription`	`string`	OPTIONAL	Detailed description	`"Help with primary school enrollment forms and fee payment"`
`serviceMode`	`enum`	REQUIRED	How the service is delivered. Values: SERVICE_DELIVERY PRODUCT_DELIVERY FACILITY_VISIT DIGITAL	`"SERVICE_DELIVERY"`
`quantityRequired`	`QuantitySpec`	OPTIONAL	What the household needs	`{"value": 2, "unit": "children", "description": "Children to enroll"}`
`beneficiaryType`	`enum`	OPTIONAL	Who the service is for. Values: INDIVIDUAL HOUSEHOLD CHILD ELDERLY	`"CHILD"`
`documentsRequired`	`string[]`	OPTIONAL	Documents the household needs to provide	`["Birth Certificate", "National ID"]`
`serviceFrequency`	`enum`	OPTIONAL	How often the service occurs. Values: ONE_TIME RECURRING SEASONAL	`"ONE_TIME"`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 language codes	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags for discoverability	`["education", "enrollment", "primary-school"]`

OpenAPI 3.1 Definition

```yaml
# HouseholdItemAttributes (hia:)
HouseholdItemAttributes:
  type: object
  required:
    - subCategory
    - serviceName
    - serviceMode
  properties:
    subCategory:
```

```yaml
        type: object
        required: [code, name]
        properties:
          code:
            type: string
            examples: ["education", "health", "insurance", "consumer-goods"]
          name:
            type: string
      serviceName:
        type: string
      serviceDescription:
        type: string
      serviceMode:
        type: string
        enum: [SERVICE_DELIVERY, PRODUCT_DELIVERY, FACILITY_VISIT, DIGITAL]
      quantityRequired:
        $ref: "#/components/schemas/QuantitySpec"
      beneficiaryType:
        type: string
        enum: [INDIVIDUAL, HOUSEHOLD, CHILD, ELDERLY]
      documentsRequired:
        type: array
        items:
          type: string
      serviceFrequency:
        type: string
        enum: [ONE_TIME, RECURRING, SEASONAL]
      availableLanguages:
        type: array
        items:
          type: string
          pattern: "^[a-z]{2}$"
      tags:
        type: array
        items:
          type: string
```

JSON Example — Beckn Message Fragment

```json
{
  "context": {
    "domain": "local-services",
    "action": "on_discover",
    "version": "2.0.0"
  },
  "message": {
    "catalog": {
      "providers": [{
        "id": "provider-uuid-020",
        "descriptor": { "name": "Kiambu Education Services" },
        "items": [{
          "id": "item-uuid-020",
          "descriptor": { "name": "School Enrollment Assistance" },
          "category_ids": ["Household-Services"],
          "itemAttributes": {
            "subCategory": { "code": "education", "name": "Education" },
            "serviceName": "School Enrollment Assistance",
            "serviceDescription": "Help with primary school enrollment forms and fee payment",
            "serviceMode": "SERVICE_DELIVERY",
            "beneficiaryType": "CHILD",
            "documentsRequired": ["Birth Certificate", "National ID"],
            "serviceFrequency": "ONE_TIME",
            "availableLanguages": ["en", "sw"],
            "tags": ["education", "enrollment"]
          }
        }]
      }]
    }
  }
}
```

3. MarketProduceItemAttributes

mpia: MarketProduceItemAttributes

Container: **itemAttributes** Applies to: **Market-Produce**

Describes the aggregation/buying service. In this flow the farmer is the seller and the provider is the buyer/aggregator.

> **Inverted roles.** Unlike other categories where the provider delivers a service to the farmer, in Market-Produce the "item" is the aggregation/buying service. The provider (aggregator) buys produce from the farmer.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code, name}	REQUIRED	Domain grouping within Market	`{"code": "livestock-products", "name": "Livestock Products"}`
`serviceName`	`string`	REQUIRED	Human-readable name	`"Egg Collection Service"`
`serviceDescription`	`string`	OPTIONAL	Detailed description	`"Weekly egg collection from farm gate"`
`produceCategory`	`string`	REQUIRED	What the aggregator buys	`"Eggs"`
`acceptedGrades`	`string[]`	OPTIONAL	Quality grades the aggregator accepts	`["Grade A", "Grade B"]`
`unitOfMeasure`	`string`	REQUIRED	Pricing/quantity unit	`"Tray (30 eggs)"`
`minimumQuantity`	`number`	OPTIONAL	Minimum lot size the aggregator will collect	`5`
`quantityAvailable`	`object` {value, unit}	OPTIONAL	What the farmer has available to sell	`{"value": 10, "unit": "Trays"}`
`collectionMode`	`enum`	REQUIRED	How produce is collected. Values: FARM_GATE_PICKUP COLLECTION_POINT FARMER_DELIVERS	`"FARM_GATE_PICKUP"`
`paymentTimeline`	`enum`	OPTIONAL	When farmer gets paid. Values: IMMEDIATE UPON_COLLECTION AFTER_SALE	`"UPON_COLLECTION"`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 language codes	`["en", "sw"]`
`tags`	`string[]`	OPTIONAL	Free-form tags	`["eggs", "poultry-products", "collection"]`

OpenAPI 3.1 Definition

```yaml
# MarketProduceItemAttributes (mpia:)
MarketProduceItemAttributes:
  type: object
  required:
    - subCategory
    - serviceName
    - produceCategory
    - unitOfMeasure
    - collectionMode
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
```

```yaml
        code:
          type: string
          examples: ["livestock-products", "farm-produce", "handicrafts"]
        name:
          type: string
    serviceName:
      type: string
    serviceDescription:
      type: string
    produceCategory:
      type: string
      description: What the aggregator buys
    acceptedGrades:
      type: array
      items:
        type: string
    unitOfMeasure:
      type: string
      description: Pricing/quantity unit
    minimumQuantity:
      type: number
      description: Minimum lot size
    quantityAvailable:
      type: object
      properties:
        value:
          type: number
        unit:
          type: string
    collectionMode:
      type: string
      enum: [FARM_GATE_PICKUP, COLLECTION_POINT, FARMER_DELIVERS]
    paymentTimeline:
      type: string
      enum: [IMMEDIATE, UPON_COLLECTION, AFTER_SALE]
    availableLanguages:
      type: array
      items:
        type: string
        pattern: "^[a-z]{2}$"
    tags:
      type: array
      items:
        type: string
```

JSON Example — Beckn Message Fragment

```json
{
  "context": {
    "domain": "local-services",
    "action": "on_discover",
    "version": "2.0.0"
  },
  "message": {
    "catalog": {
      "providers": [{
        "id": "provider-uuid-030",
        "descriptor": { "name": "Kienyeji Egg Traders Ltd" },
        "items": [{
          "id": "item-uuid-030",
          "descriptor": { "name": "Egg Collection Service" },
          "category_ids": ["Market-Produce"],
          "itemAttributes": {
            "subCategory": { "code": "livestock-products", "name": "Livestock Products" },
            "serviceName": "Egg Collection Service",
            "produceCategory": "Eggs",
            "acceptedGrades": ["Grade A", "Grade B"],
            "unitOfMeasure": "Tray (30 eggs)",
            "minimumQuantity": 5,
            "quantityAvailable": { "value": 10, "unit": "Trays" },
            "collectionMode": "FARM_GATE_PICKUP",
            "paymentTimeline": "UPON_COLLECTION",
            "tags": ["eggs", "poultry-products"]
```

```
          }
        }]
      }]
    }
  }
}
```

4. AdvisoryContentItemAttributes

acia: AdvisoryContentItemAttributes

Container: **itemAttributes** Applies to: **Advisory-Content**

Describes free advisory/training content. No order lifecycle — content URL is delivered directly in `on_discover`.

> **Advisory is discover-only.** There is no select/init/confirm flow for advisory content. The `contentUrl` is delivered directly in the `on_discover` response, and no `offerAttributes` or `orderAttributes` are used.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`subCategory`	`object` {code, name}	REQUIRED	Topic grouping within Advisory	`{"code": "agricultural-advisory", "name": "Agricultural Advisory"}`
`serviceName`	`string`	REQUIRED	Content title	`"Broiler Feeding Schedule - Week 1"`
`serviceDescription`	`string`	OPTIONAL	Content description	`"Optimal feeding schedule and quantities for week-1 broiler chicks"`
`contentType`	`enum`	REQUIRED	Media type. Values: TEXT VIDEO PDF AUDIO IMAGE INFOGRAPHIC	`"VIDEO"`
`contentMimeType`	`string`	OPTIONAL	IANA MIME type	`"video/mp4"`
`contentUrl`	`string` (uri)	REQUIRED	Direct URL to the content	`"https://platformcdn.kuza.one/global/content/video/feeding_schedule.mp4"`
`contentDuration`	`string` ISO 8601	OPTIONAL	Duration for video/audio content	`"PT3M45S"`
`targetAudience`	`string`	OPTIONAL	Who the content is most relevant for	`"Poultry farmers in first cycle"`
`availableLanguages`	`string[]`	OPTIONAL	ISO 639-1 language codes	`["en", "sw"]`

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`tags`	`string[]`	OPTIONAL	Free-form tags	`["feeding", "broiler", "week-1"]`

OpenAPI 3.1 Definition

```yaml
# AdvisoryContentItemAttributes (acia:)
AdvisoryContentItemAttributes:
  type: object
  required:
    - subCategory
    - serviceName
    - contentType
    - contentUrl
  properties:
    subCategory:
      type: object
      required: [code, name]
      properties:
        code:
          type: string
          examples: ["agricultural-advisory", "financial-literacy", "health-nutrition"]
        name:
          type: string
    serviceName:
      type: string
      description: Content title
    serviceDescription:
      type: string
    contentType:
      type: string
      enum: [TEXT, VIDEO, PDF, AUDIO, IMAGE, INFOGRAPHIC]
    contentMimeType:
      type: string
      description: IANA MIME type
    contentUrl:
      type: string
      format: uri
      description: Direct URL to content
    contentDuration:
      type: string
      format: duration
      description: ISO 8601 duration for video/audio
    targetAudience:
      type: string
    availableLanguages:
      type: array
      items:
        type: string
        pattern: "^[a-z]{2}$"
    tags:
      type: array
      items:
        type: string
```

JSON Example — Beckn Message Fragment

```json
{
  "context": {
    "domain": "local-services",
    "action": "on_discover",
    "version": "2.0.0"
  },
  "message": {
    "catalog": {
      "providers": [{
        "id": "provider-uuid-kuza",
        "descriptor": { "name": "Kuza Advisory Platform" },
        "items": [{
          "id": "item-uuid-040",
```

```
            "descriptor": { "name": "Broiler Feeding Schedule - Week 1" },
            "category_ids": ["Advisory-Content"],
            "itemAttributes": {
              "subCategory": { "code": "agricultural-advisory", "name": "Agricultural Advisory" },
              "serviceName": "Broiler Feeding Schedule - Week 1",
              "serviceDescription": "Optimal feeding schedule and quantities for week-1 broiler chicks",
              "contentType": "VIDEO",
              "contentMimeType": "video/mp4",
              "contentUrl": "https://platformcdn.kuza.one/global/content/video/feeding_wk1.mp4",
              "contentDuration": "PT3M45S",
              "targetAudience": "Poultry farmers in first cycle",
              "availableLanguages": ["en", "sw"],
              "tags": ["feeding", "broiler", "week-1"]
            }
          }]
        }]
      }
    }
  }
```

5. LocalServiceOfferAttributes

lsoa: LocalServiceOfferAttributes

Container: **offerAttributes** Applies to: **Livelihood-Services, Household-Services, Market-Produce**

Pricing, payment methods, availability, and capacity for all paid flows. Not used for Advisory-Content.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`priceAmount`	`number`	REQUIRED	Base price	`50`
`currency`	`string`	REQUIRED	ISO 4217 currency code	`"KES"`
`priceUnit`	`string`	OPTIONAL	What the price is per	`"per bird"`
`pricingModel`	`enum`	REQUIRED	Pricing approach. Values: FIXED PER_UNIT PER_VISIT NEGOTIABLE	`"PER_UNIT"`
`acceptedPaymentMethods`	`enum[]`	REQUIRED	How the provider accepts payment. Values: CASH MPESA	`["CASH", "MPESA"]`
`capacity`	`object`	OPTIONAL	Provider capacity: {value, unit, period}	`{"value": 500, "unit": "birds", "period": "per day"}`
`serviceAvailability`	`object`	OPTIONAL	When the provider is available: {availableDays, timeSlots}	`{"availableDays": ["Mon","Tue","Wed","Thu","Fri"], "timeSlots": ["08:00-12:00","14:00-17:00"]}`
`cancellationPolicy`	`object`	OPTIONAL	Cancellation rules: {isCancellable, windowHours, reasons}	`{"isCancellable": true, "windowHours": 24, "reasons": ["Weather","Emergency"]}`
`validFrom`	`string` (datetime)	OPTIONAL	Offer start date (ISO 8601)	`"2026-04-01T00:00:00Z"`
`validTo`	`string` (datetime)	OPTIONAL	Offer expiry date (ISO 8601)	`"2026-06-30T23:59:59Z"`

OpenAPI 3.1 Definition

```yaml
# LocalServiceOfferAttributes (lsoa:)
LocalServiceOfferAttributes:
  type: object
  required:
    - priceAmount
    - currency
    - pricingModel
    - acceptedPaymentMethods
  properties:
    priceAmount:
      type: number
      description: Base price
    currency:
      type: string
      pattern: "^[A-Z]{3}$"
      description: ISO 4217 currency code
    priceUnit:
      type: string
      description: What the price is per (e.g., "per bird", "per visit")
    pricingModel:
      type: string
      enum: [FIXED, PER_UNIT, PER_VISIT, NEGOTIABLE]
    acceptedPaymentMethods:
      type: array
      items:
        type: string
        enum: [CASH, MPESA]
      minItems: 1
    capacity:
      type: object
      properties:
        value:
          type: number
        unit:
          type: string
        period:
          type: string
    serviceAvailability:
      type: object
      properties:
        availableDays:
          type: array
          items:
            type: string
        timeSlots:
          type: array
          items:
            type: string
    cancellationPolicy:
      type: object
      properties:
        isCancellable:
          type: boolean
        windowHours:
          type: integer
        reasons:
          type: array
          items:
            type: string
    validFrom:
      type: string
      format: date-time
    validTo:
      type: string
      format: date-time
```

JSON Example — Beckn Message Fragment

```json
{
  "offerAttributes": {
    "priceAmount": 50,
    "currency": "KES",
```

```
    "priceUnit": "per bird",
    "pricingModel": "PER_UNIT",
    "acceptedPaymentMethods": ["CASH", "MPESA"],
    "capacity": { "value": 500, "unit": "birds", "period": "per day" },
    "serviceAvailability": {
      "availableDays": ["Mon", "Tue", "Wed", "Thu", "Fri"],
      "timeSlots": ["08:00-12:00", "14:00-17:00"]
    },
    "cancellationPolicy": {
      "isCancellable": true,
      "windowHours": 24,
      "reasons": ["Weather", "Emergency"]
    },
    "validFrom": "2026-04-01T00:00:00Z",
    "validTo": "2026-06-30T23:59:59Z"
  }
}
```

6. LocalServiceFulfillmentAttributes

lsfa: LocalServiceFulfillmentAttributes

Container: **fulfillmentAttributes** Applies to: **All flows** (including Advisory for location context)

Location, scheduling, delivery mode, and service status tracking. Used across all four category flows.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`serviceArea`	`ServiceArea[]`	REQUIRED (PROVIDER)	Provider's registered coverage areas	`[{"countyCode": "022", "countyName": "Kiambu", "subCounties": [{"subCountyCode": "022-01", "subCountyName": "Thika Town"}]}]`
`farmerLocation`	`FarmerLocation`	REQUIRED (FARMER)	Farmer's location details	`{"county": "Kiambu", "subCounty": "Thika Town", "ward": "Kamenu", "village": "Makongeni"}`
`scheduledDate`	`string` (date)	OPTIONAL	Farmer's preferred service date (ISO 8601)	`"2026-04-15"`
`scheduledTimeSlot`	`string`	OPTIONAL	Preferred time of day	`"Morning"`
`confirmedVisitTime`	`string` (datetime)	OPTIONAL	Provider's confirmed visit date/time	`"2026-04-15T09:00:00+03:00"`
`actualServiceDate`	`string` (date)	OPTIONAL	When service was actually provided	`"2026-04-15"`
`deliveryMode`	`enum`	REQUIRED	How the service is delivered. Values: ON_SITE PROVIDER_LOCATION COLLECTION_POINT DIGITAL	`"ON_SITE"`
`serviceStatus`	`enum`	REQUIRED	Current status in lifecycle. Values: ORDER_PLACED PROVIDER_ACCEPTED PROVIDER_REJECTED IN_PROGRESS COMPLETED FARMER_CONFIRMED CANCELLED CLOSED	`"PROVIDER_ACCEPTED"`
`providerNotes`	`string`	OPTIONAL	Notes from the provider's visit	`"Vaccinated 195 birds. 5 birds showed signs of illness, recommended`

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
				`isolation."`
`followUpRequired`	`boolean`	OPTIONAL	Provider flags if follow-up is needed	`true`

OpenAPI 3.1 Definition

```yaml
# LocalServiceFulfillmentAttributes (lsfa:)
LocalServiceFulfillmentAttributes:
  type: object
  required:
    - deliveryMode
    - serviceStatus
  properties:
    serviceArea:
      type: array
      description: Provider's registered coverage areas (required for provider)
      items:
        $ref: "#/components/schemas/ServiceArea"
    farmerLocation:
      $ref: "#/components/schemas/FarmerLocation"
    scheduledDate:
      type: string
      format: date
      description: Farmer's preferred service date
    scheduledTimeSlot:
      type: string
      description: Preferred time (e.g., "Morning", "Afternoon")
    confirmedVisitTime:
      type: string
      format: date-time
      description: Provider's confirmed date/time
    actualServiceDate:
      type: string
      format: date
      description: When service was actually provided
    deliveryMode:
      type: string
      enum: [ON_SITE, PROVIDER_LOCATION, COLLECTION_POINT, DIGITAL]
    serviceStatus:
      type: string
      enum:
        - ORDER_PLACED
        - PROVIDER_ACCEPTED
        - PROVIDER_REJECTED
        - IN_PROGRESS
        - COMPLETED
        - FARMER_CONFIRMED
        - CANCELLED
        - CLOSED
    providerNotes:
      type: string
    followUpRequired:
      type: boolean
```

JSON Example — Beckn Message Fragment

```json
{
  "fulfillmentAttributes": {
    "serviceArea": [{
      "countyCode": "022",
      "countyName": "Kiambu",
      "subCounties": [{
        "subCountyCode": "022-01",
        "subCountyName": "Thika Town"
      }]
    }],
    "farmerLocation": {
      "county": "Kiambu",
      "subCounty": "Thika Town",
```

```
        "ward": "Kamenu",
        "village": "Makongeni",
        "gpsCoordinates": { "latitude": -1.0396, "longitude": 37.0900 }
      },
      "scheduledDate": "2026-04-15",
      "scheduledTimeSlot": "Morning",
      "confirmedVisitTime": "2026-04-15T09:00:00+03:00",
      "deliveryMode": "ON_SITE",
      "serviceStatus": "PROVIDER_ACCEPTED",
      "followUpRequired": false
    }
  }
```

7. LocalServiceOrderAttributes

lsor: LocalServiceOrderAttributes

Container: **orderAttributes** Applies to: **Livelihood-Services, Household-Services, Market-Produce**

Post-transaction attributes for all paid flows. Both sides (farmer and provider) report payment status independently, enabling reconciliation without a shared payment gateway.

Dual-sided reporting. Payment is reported independently by both parties — the farmer reports via WhatsApp and the provider reports via the web app. There is no integrated payment gateway; reconciliation happens by comparing both reports.

Field Reference

FIELD	TYPE	REQUIRED	DESCRIPTION	EXAMPLE
`farmerPaymentMethod`	`enum`	OPTIONAL	How the farmer paid. Values: CASH MPESA	`"MPESA"`
`farmerPaymentStatus`	`enum`	OPTIONAL	Farmer's self-reported payment status. Values: PAID NOT_PAID	`"PAID"`
`providerPaymentReceived`	`boolean`	OPTIONAL	Provider confirms receipt of payment	`true`
`providerAmountReceived`	`number`	OPTIONAL	Amount the provider received	`10000`
`paymentCurrency`	`string`	OPTIONAL	ISO 4217 currency code	`"KES"`
`serviceConfirmedByFarmer`	`boolean`	OPTIONAL	Farmer confirms the service was performed	`true`
`providerFulfillmentStatus`	`enum`	OPTIONAL	Provider's self-reported fulfillment status. Values: COMPLETED NOT_COMPLETED	`"COMPLETED"`
`providerFeedback`	`Feedback`	OPTIONAL	Provider's feedback: {rating, selectedOption, additionalText}	`{"rating": 4, "selectedOption": "Good experience", "additionalText": "Farmer was prepared"}`
`farmerFeedback`	`Feedback`	OPTIONAL	Farmer's feedback: {rating, selectedOption, additionalText}	`{"rating": 5, "selectedOption": "Excellent service"}`
`followUpRequired`	`boolean`	OPTIONAL	Provider flags if follow-up is needed	`true`
`followUpNotes`	`string`	OPTIONAL	What follow-up is needed	`"Booster vaccination needed in 2 weeks"`

OpenAPI 3.1 Definition

```yaml
# LocalServiceOrderAttributes (lsor:)
LocalServiceOrderAttributes:
  type: object
  properties:
    farmerPaymentMethod:
      type: string
      enum: [CASH, MPESA]
      description: Reported by farmer via WhatsApp
    farmerPaymentStatus:
      type: string
      enum: [PAID, NOT_PAID]
    providerPaymentReceived:
      type: boolean
      description: Provider confirms payment receipt via App
    providerAmountReceived:
      type: number
    paymentCurrency:
      type: string
      pattern: "^[A-Z]{3}$"
    serviceConfirmedByFarmer:
      type: boolean
      description: Farmer confirms service was done
    providerFulfillmentStatus:
      type: string
      enum: [COMPLETED, NOT_COMPLETED]
    providerFeedback:
      $ref: "#/components/schemas/Feedback"
    farmerFeedback:
      $ref: "#/components/schemas/Feedback"
    followUpRequired:
      type: boolean
    followUpNotes:
      type: string
```

JSON Example — Beckn Message Fragment

```json
{
  "orderAttributes": {
    "farmerPaymentMethod": "MPESA",
    "farmerPaymentStatus": "PAID",
    "providerPaymentReceived": true,
    "providerAmountReceived": 10000,
    "paymentCurrency": "KES",
    "serviceConfirmedByFarmer": true,
    "providerFulfillmentStatus": "COMPLETED",
    "providerFeedback": {
      "rating": 4,
      "selectedOption": "Good experience",
      "additionalText": "Farmer was well-prepared, birds accessible"
    },
    "farmerFeedback": {
      "rating": 5,
      "selectedOption": "Excellent service"
    },
    "followUpRequired": true,
    "followUpNotes": "Booster vaccination needed in 2 weeks"
  }
}
```

Sub-schemas

These shared types are referenced across multiple extension schemas.

ServiceArea

Represents a geographic coverage area used by providers to define where they operate.

FIELD	TYPE	REQUIRED	DESCRIPTION
`countyCode`	`string`	REQUIRED	County identifier code
`countyName`	`string`	REQUIRED	County name
`subCounties`	`object[]`	OPTIONAL	Array of {subCountyCode, subCountyName}

```yaml
# ServiceArea sub-schema
ServiceArea:
  type: object
  required: [countyCode, countyName]
  properties:
    countyCode:
      type: string
    countyName:
      type: string
    subCounties:
      type: array
      items:
        type: object
        required: [subCountyCode, subCountyName]
        properties:
          subCountyCode:
            type: string
          subCountyName:
            type: string
```

FarmerLocation

Represents the farmer's geographic location for service matching and fulfillment.

FIELD	TYPE	REQUIRED	DESCRIPTION
`county`	`string`	REQUIRED	County name
`subCounty`	`string`	REQUIRED	Sub-county name
`ward`	`string`	OPTIONAL	Ward name
`village`	`string`	OPTIONAL	Village name
`gpsCoordinates`	`object`	OPTIONAL	{latitude: number, longitude: number}

```yaml
# FarmerLocation sub-schema
FarmerLocation:
  type: object
  required: [county, subCounty]
  properties:
    county:
      type: string
    subCounty:
      type: string
    ward:
      type: string
    village:
      type: string
    gpsCoordinates:
      type: object
      properties:
        latitude:
          type: number
        longitude:
          type: number
```

QuantitySpec

A generic quantity descriptor used across item attributes to specify amounts.

FIELD	TYPE	REQUIRED	DESCRIPTION
`value`	`number`	OPTIONAL	Numeric quantity
`unit`	`string`	OPTIONAL	Unit of measure
`description`	`string`	OPTIONAL	Human-readable description

```yaml
# QuantitySpec sub-schema
QuantitySpec:
  type: object
  properties:
    value:
      type: number
    unit:
      type: string
    description:
      type: string
```

Feedback

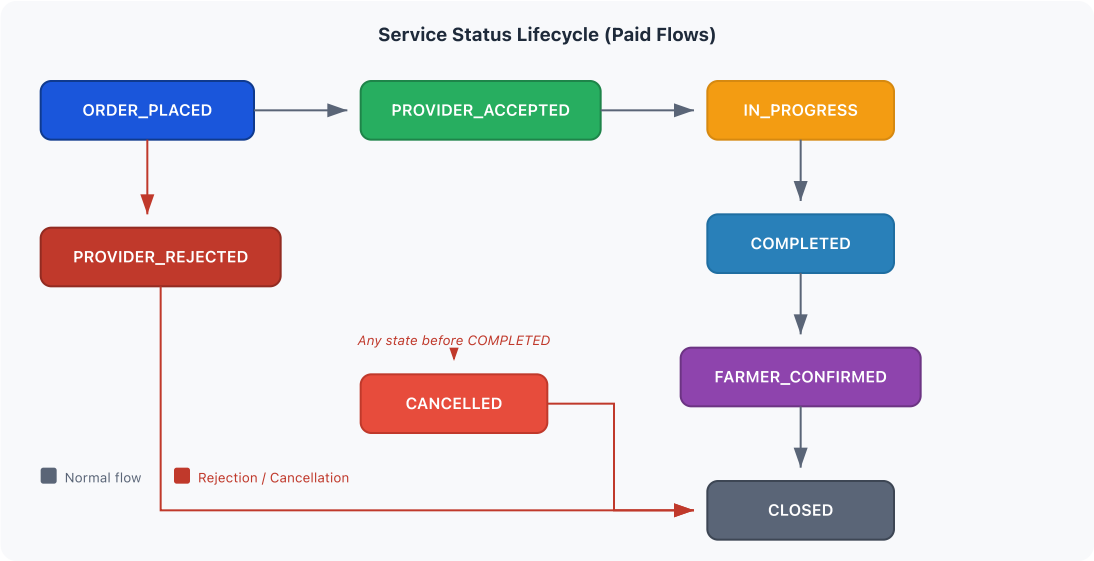
Feedback from either party (farmer or provider) after service completion.

FIELD	TYPE	REQUIRED	DESCRIPTION
`rating`	`integer`	OPTIONAL	Rating from 1 to 5
`selectedOption`	`string`	OPTIONAL	Pre-defined feedback option
`additionalText`	`string`	OPTIONAL	Free-text additional feedback

```yaml
# Feedback sub-schema
Feedback:
  type: object
  properties:
    rating:
      type: integer
      minimum: 1
      maximum: 5
    selectedOption:
      type: string
    additionalText:
      type: string
```

Service Status Lifecycle

The `serviceStatus` field in `LocalServiceFulfillmentAttributes` tracks the order through the following state machine. Advisory content has no lifecycle — it is delivered immediately via `contentUrl` in `on_discover`.



State transition rules:

FROM	TO	TRIGGERED BY
`ORDER_PLACED`	`PROVIDER_ACCEPTED`	Provider accepts the order via App
`ORDER_PLACED`	`PROVIDER_REJECTED`	Provider rejects the order via App
`PROVIDER_ACCEPTED`	`IN_PROGRESS`	Provider begins service delivery
`IN_PROGRESS`	`COMPLETED`	Provider marks service as done via App
`COMPLETED`	`FARMER_CONFIRMED`	Farmer confirms service was received via WhatsApp
`FARMER_CONFIRMED`	`CLOSED`	Both parties have reported; transaction finalized
`PROVIDER_REJECTED`	`CLOSED`	Automatic closure after rejection
Any state before `COMPLETED`	`CANCELLED`	Either party cancels
`CANCELLED`	`CLOSED`	Automatic closure after cancellation

> **Advisory-Content has no lifecycle.** Content is delivered immediately via `contentUrl` in the `on_discover` response. No status tracking is needed.

Full OpenAPI 3.1 Bundle

The complete schema definitions bundled as a single OpenAPI 3.1 components block, suitable for inclusion in a Beckn domain specification file.

```yaml
# ============================================================
# NFH-Kuza-Google Extension Schemas
# Domain: local-services | Protocol: Beckn v2.0.0
# ============================================================

openapi: "3.1.0"
info:
  title: NFH-Kuza-Google Extension Schemas
  version: "1.0.0"
  description: >
    Domain extension schemas for the local-services domain
    in the NFH-Kuza-Google rural service delivery pilot.

components:
  schemas:

    # ---- Sub-schemas ----
```

```yaml
    QuantitySpec:
      type: object
      properties:
        value:
          type: number
        unit:
          type: string
        description:
          type: string

    ServiceArea:
      type: object
      required: [countyCode, countyName]
      properties:
        countyCode:
          type: string
        countyName:
          type: string
        subCounties:
          type: array
          items:
            type: object
            required: [subCountyCode, subCountyName]
            properties:
              subCountyCode:
                type: string
              subCountyName:
                type: string

    FarmerLocation:
      type: object
      required: [county, subCounty]
      properties:
        county:
          type: string
        subCounty:
          type: string
        ward:
          type: string
        village:
          type: string
        gpsCoordinates:
          type: object
          properties:
            latitude:
              type: number
            longitude:
              type: number

    Feedback:
      type: object
      properties:
        rating:
          type: integer
          minimum: 1
          maximum: 5
        selectedOption:
          type: string
        additionalText:
          type: string

    # ---- Item Attributes ----

    LivelihoodItemAttributes:
      type: object
      required: [subCategory, serviceName, serviceMode]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        serviceName: { type: string }
```

```yaml
        serviceDescription: { type: string }
        serviceMode:
          type: string
          enum: [ON_SITE_VISIT, PROVIDER_LOCATION, INPUT_DELIVERY]
        quantityRequired: { $ref: "#/components/schemas/QuantitySpec" }
        estimatedDuration: { type: string, format: duration }
        inputDetails:
          type: object
          properties:
            inputName: { type: string }
            unit: { type: string }
            quantityPerUnit: { type: number }
        sectorContext:
          type: object
          properties:
            livestockType: { type: string }
            breed: { type: string }
            cropType: { type: string }
            tradeType: { type: string }
        availableLanguages:
          type: array
          items: { type: string, pattern: "^[a-z]{2}$" }
        tags:
          type: array
          items: { type: string }

      HouseholdItemAttributes:
        type: object
        required: [subCategory, serviceName, serviceMode]
        properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        serviceName: { type: string }
        serviceDescription: { type: string }
        serviceMode:
          type: string
          enum: [SERVICE_DELIVERY, PRODUCT_DELIVERY, FACILITY_VISIT, DIGITAL]
        quantityRequired: { $ref: "#/components/schemas/QuantitySpec" }
        beneficiaryType:
          type: string
          enum: [INDIVIDUAL, HOUSEHOLD, CHILD, ELDERLY]
        documentsRequired:
          type: array
          items: { type: string }
        serviceFrequency:
          type: string
          enum: [ONE_TIME, RECURRING, SEASONAL]
        availableLanguages:
          type: array
          items: { type: string, pattern: "^[a-z]{2}$" }
        tags:
          type: array
          items: { type: string }

      MarketProduceItemAttributes:
        type: object
        required: [subCategory, serviceName, produceCategory, unitOfMeasure, collectionMode]
        properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        serviceName: { type: string }
        serviceDescription: { type: string }
        produceCategory: { type: string }
        acceptedGrades:
          type: array
          items: { type: string }
        unitOfMeasure: { type: string }
```

```yaml
      minimumQuantity: { type: number }
      quantityAvailable:
        type: object
        properties:
          value: { type: number }
          unit: { type: string }
      collectionMode:
        type: string
        enum: [FARM_GATE_PICKUP, COLLECTION_POINT, FARMER_DELIVERS]
      paymentTimeline:
        type: string
        enum: [IMMEDIATE, UPON_COLLECTION, AFTER_SALE]
      availableLanguages:
        type: array
        items: { type: string, pattern: "^[a-z]{2}$" }
      tags:
        type: array
        items: { type: string }

    AdvisoryContentItemAttributes:
      type: object
      required: [subCategory, serviceName, contentType, contentUrl]
      properties:
        subCategory:
          type: object
          required: [code, name]
          properties:
            code: { type: string }
            name: { type: string }
        serviceName: { type: string }
        serviceDescription: { type: string }
        contentType:
          type: string
          enum: [TEXT, VIDEO, PDF, AUDIO, IMAGE, INFOGRAPHIC]
        contentMimeType: { type: string }
        contentUrl: { type: string, format: uri }
        contentDuration: { type: string, format: duration }
        targetAudience: { type: string }
        availableLanguages:
          type: array
          items: { type: string, pattern: "^[a-z]{2}$" }
        tags:
          type: array
          items: { type: string }

    # ---- Offer Attributes ----

    LocalServiceOfferAttributes:
      type: object
      required: [priceAmount, currency, pricingModel, acceptedPaymentMethods]
      properties:
        priceAmount: { type: number }
        currency: { type: string, pattern: "^[A-Z]{3}$" }
        priceUnit: { type: string }
        pricingModel:
          type: string
          enum: [FIXED, PER_UNIT, PER_VISIT, NEGOTIABLE]
        acceptedPaymentMethods:
          type: array
          items: { type: string, enum: [CASH, MPESA] }
          minItems: 1
        capacity:
          type: object
          properties:
            value: { type: number }
            unit: { type: string }
            period: { type: string }
        serviceAvailability:
          type: object
          properties:
            availableDays:
              type: array
              items: { type: string }
            timeSlots:
              type: array
```

```yaml
            items: { type: string }
      cancellationPolicy:
        type: object
        properties:
          isCancellable: { type: boolean }
          windowHours: { type: integer }
          reasons:
            type: array
            items: { type: string }
      validFrom: { type: string, format: date-time }
      validTo: { type: string, format: date-time }

    # ---- Fulfillment Attributes ----

    LocalServiceFulfillmentAttributes:
      type: object
      required: [deliveryMode, serviceStatus]
      properties:
        serviceArea:
          type: array
          items: { $ref: "#/components/schemas/ServiceArea" }
        farmerLocation:
          $ref: "#/components/schemas/FarmerLocation"
        scheduledDate: { type: string, format: date }
        scheduledTimeSlot: { type: string }
        confirmedVisitTime: { type: string, format: date-time }
        actualServiceDate: { type: string, format: date }
        deliveryMode:
          type: string
          enum: [ON_SITE, PROVIDER_LOCATION, COLLECTION_POINT, DIGITAL]
        serviceStatus:
          type: string
          enum:
            - ORDER_PLACED
            - PROVIDER_ACCEPTED
            - PROVIDER_REJECTED
            - IN_PROGRESS
            - COMPLETED
            - FARMER_CONFIRMED
            - CANCELLED
            - CLOSED
        providerNotes: { type: string }
        followUpRequired: { type: boolean }

    # ---- Order Attributes ----

    LocalServiceOrderAttributes:
      type: object
      properties:
        farmerPaymentMethod:
          type: string
          enum: [CASH, MPESA]
        farmerPaymentStatus:
          type: string
          enum: [PAID, NOT_PAID]
        providerPaymentReceived: { type: boolean }
        providerAmountReceived: { type: number }
        paymentCurrency: { type: string, pattern: "^[A-Z]{3}$" }
        serviceConfirmedByFarmer: { type: boolean }
        providerFulfillmentStatus:
          type: string
          enum: [COMPLETED, NOT_COMPLETED]
        providerFeedback: { $ref: "#/components/schemas/Feedback" }
        farmerFeedback: { $ref: "#/components/schemas/Feedback" }
        followUpRequired: { type: boolean }
        followUpNotes: { type: string }
```